

K-Bound: When Is Label-Free Adaptation Knowable?

Pratik Niroula

Minnesota State University, Mankato

pratik.niroula@mnsu.edu

Abstract

Test-time adaptation can improve or degrade a model, leaving a separate decision: when does label-free evidence justify replacing the frozen predictor? K-Bound characterizes this decision in a binary population model under an externally declared calibration-residual bound β . Holding label-free observations fixed, we derive the exact feasible interval for paired benefit: a strict decision is justified precisely when $|M| > \beta$ on disagreement. An evidence-fibre audit shows why the same observations cannot justify a smaller uniform residual bound without additional structure. A separate finite-sample multiclass construction bounds paired benefit under an explicit conditional-transport budget; synthetic examples illustrate its value and failure under assumption violations. Knowability-Guided Adaptation (KGA) is an empirical companion, not an implementation of the population frontier. Using labeled historical outcomes, it predicts observed evaluation-cell benefit and chooses ADAPT, FREEZE, or ABSTAIN through residual intervals. Controlled studies show candidate-dependent routing gains, while SAR favors always-adapt. A shared-predictor comparison finds lower observed loss for CIFAR Tent at a prespecified utility-harm trade-off; this advantage is smaller at matched exposure and depends on the corruption mix. Simpler rules often win. Nominal 90% constant-cell inclusion is only 50.51–60.69% on held-out CIFAR corruptions. CCT-20 has no ADAPT decisions, and So2Sat stops before target scoring. These findings do not establish interval coverage or population-risk protection on unseen natural shifts; useful prospective routing remains open.

Keywords: test-time adaptation; abstention; distribution shift; label-free risk estimation; conformal calibration; selective decisions.

1 Introduction

An update that improves a model on one deployment distribution can degrade it on another. Test-time adaptation proposes updates using unlabeled

data through entropy minimization, filtering, regularization, and related methods [1, 4, 5, 6]. Choosing whether to serve the resulting candidate is a separate decision. The challenge is to justify that choice when the target labels needed to measure its benefit are unavailable.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

A positive benefit favors the update; a negative benefit favors the frozen model. We distinguish three conclusions: ADAPT supports positive benefit, FREEZE supports negative benefit, and ABSTAIN records insufficient evidence for either strict direction. Both non-adapt actions retain f_0 , but only freeze asserts that the candidate is worse.

K-Bound studies this decision as a partial-identification problem. In a binary model, we hold the input law, predictors, and label-free evidence fixed and vary the target labels within a declared class. The resulting feasible benefit set gives an exact boundary for a uniformly justified strict decision. The bound on the hidden calibration residual is an assumption about that class, not a quantity that an unlabeled batch supplies for free. A second result makes this information limit explicit: an audit using the same evidence must still cover the unresolved residual range.

The theory asks what a fixed label-free evidence law and a declared residual bound can identify. Knowability-Guided Adaptation (KGA) studies the same update decision using additional information: labeled historical shift outcomes. It fits an empirical benefit predictor and calibrates a three-way decision rule. The connection is the decision problem; the empirical gate does not implement the population frontier. Section 6.1 gives the full relationship between their targets and assumptions. Together, these analyses separate two possible obstacles to a justified update: the permitted evidence may leave its benefit unidentified, or an empirical interval may fail to

transfer. The first requires additional assumptions or information; the second requires evidence for the calibration procedure at the intended decision unit.

1.1 Three Scientific Contributions

1. **An exact benefit frontier on a declared class.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **An evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **An empirical paired-benefit gate.** We give a reproducible three-way interval procedure and evaluate its commitment behavior under controlled and natural shifts. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The empirical studies then ask whether an interval improves the update decision and whether its calibration transfers beyond the fitting conditions. Controlled CIFAR-10-C results are candidate-dependent: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt. A comparison using a shared point predictor isolates the contribution of calibration from the predictor itself (Section 8.7). It finds a benefit for CIFAR Tent at one harm-weighted operating point, but simple rules win elsewhere and nominal interval inclusion deteriorates on held-out corruptions.

Natural-shift studies expose a further distinction between retaining utility and selecting useful updates. On CCT-20, KGA retained the frozen model in all 45 cells, avoiding the measured loss of always-adapt while making no ADAPT decisions. The So2Sat studies stopped before target scoring. These findings leave useful prospective routing unresolved. A separate paired-transport construction shows how a declared conditional-shift assumption can support a finite-sample multiclass decision, with synthetic examples illustrating its value and its failure when that assumption is violated. Additional adverse and incomplete studies are reported in the supplement.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training improve candidate construction through entropy objectives, sample filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks [1, 4, 5, 6, 7, 2, 3]. Selective methods also choose where or when to update. TTSA selects modality-specific adapters [10]; GALA selects layers and filters samples using gradient alignment [11]; Informed Adaptation learns topological features associated with useful updates [9]. KGA evaluates a fixed proposed candidate against its frozen baseline. Learned routing and rollback are established ideas; their existence is not our novelty claim. MORPHEUS [58], presented at the ICLR 2026 Test-Time Updates workshop, learns historical regressors from source-model entropy and embedding geometry to predict post-adaptation accuracy and choose a method before executing it. This directly overlaps the practical motivation of avoiding harmful updates. KGA’s proposed distinction is the calibrated paired-benefit operating point for one executed shadow candidate; whether that calibration improves useful decisions over a shared point predictor is an empirical question evaluated separately.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information [15, 17, 16, 18, 35, 34]. The closest empirical comparison is Kim et al.’s NeurIPS 2024 work, *Test-Time Adaptation Induces Stronger Accuracy and Agreement-on-the-Line* [51]. It uses agreement-based ALine estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA instead fits the paired observed benefit of one frozen/candidate pair and uses an interval to make a three-way decision. This is a difference in the fitted target and decision procedure, not a claim that accuracy estimates cannot be differenced to compare models. In Section 8.3 (Table 9), we compare current KGA with fixed policies, scalar evidence rules and a point gate sharing its exact predictions. These saved-feature comparisons isolate decision rules; they do not establish equivalence to complete published TTA/ALine, AETTA, or POEM systems. Method-

KGA decision flow

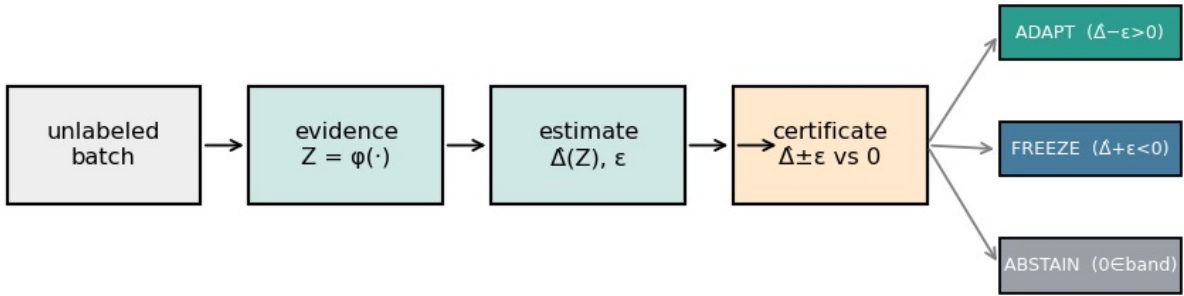


Figure 1: KGA’s empirical workflow. A shadow candidate proposes an update; a learned benefit prediction and residual interval determine whether to accept it. Historical labels support development and calibration, while the new decision uses no evaluation labels.

specific implementation and calibration qualifications are given in Appendix B.3.4.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting [14]. Schirmer et al. monitor the running risk of an adapting model relative to a source-risk reference, under a proxy-informativeness condition [8]; the supervised precursor uses a labeled test stream [13]. Drift-to-Action Controllers use delayed labels and online certificates to gate adaptation, abstention, rollback, and retraining [50]. These methods differ in the risk target, label access, and sequential guarantee. KGA concerns one paired benefit and does not establish an anytime-valid risk guarantee.

2.4 Partial Identification and Robust Decisions

An identified set describes the values compatible with the observations and assumptions. Decision theory under partial identification studies how to act when that set leaves payoffs ambiguous; Christensen et al. combine worst-case payoff uncertainty with statistical learning of identified parameters [54]. K-Bound addresses the narrower question of a uniformly correct strict benefit sign. It does not solve their general optimal-decision problem or claim that interval thresholding itself is new.

Indistinguishable adaptation problems are already central to domain-adaptation impossibility [20, 49]. K-Bound’s specific contribution is a disagreement-

conditional label-kernel construction, its clipped benefit set, and the associated residual-audit floor. The proof does not introduce label-free impossibility. The study in [56] independently examines when an unlabeled observation channel identifies the best TTA action. That September 2026 preprint includes a finite-batch Gaussian keep/recenter boundary and benchmark examples where stream structure changes the preferred action. Thus action identification being weaker than mechanism identification is not a unique claim of this paper. Our population result fixes a realized predictor pair and a disagreement-conditional residual class, deriving its clipped paired-benefit set and strict-sign boundary. The distinction lies in that explicit compatible-law construction and its audit requirements, not in the general fibre-consistency principle. Risk-estimation assumptions based on conditional independence [19], label shift [44, 45], or disagreement discrepancy [21] restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure [12]. Such restrictions can change the identified set; they are not contradicted by a converse theorem for a richer class. The construction in [57] obtains finite-sample label-shift confidence regions through matrix constraints and linear programming. Consequently, optimizing a compatible probability table is established machinery. The separate paired-transport extension evaluated here targets the benefit contrast directly, retains joint prediction-pair information, and reports sensitivity to an externally specified conditional-transport violation. It does not estimate that violation from unlabeled data or claim a new general-purpose LP confidence

method.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions [22, 46, 47]. KGA abstains on an update. Its coverage-to-action proposition applies standard interval logic [36, 37, 40, 41]. The main statistical work is to justify coverage, especially after selection or under shift.

Jin and Ren’s JOMI procedure targets coverage conditional on a unit being selected, under exchangeability and a compatible selection rule [52]. Gibbs et al. provide conditional guarantees over specified classes of shifts using a different calibration construction [53]. KGA’s ordinary residual interval implements neither procedure. Its marginal false-adapt bound therefore must be distinguished from error conditional on ADAPT and from coverage within each target environment. Learn Then Test separately addresses configuration selection and multiplicity [41].

3 Setup, Notation, and Claim Levels

3.1 Model Pair, Evidence, and Evaluation Unit

Let P_S and P_T be the source and target laws and $R_T(f) = \mathbb{E}_{(X,Y) \sim P_T}[\ell(f(X), Y)]$. Population risks integrate a fresh draw from P_T , holding the realized model pair fixed. An adapter $\mathcal{A}$ proposes f_a from f_0 and unlabeled inputs $X_{1:m}$. The decision can read

$$Z = \phi(X_{1:m}, f_0, f_a),$$

including predictions, logits, and update statistics, but not the new evaluation labels.

For evaluation unit j , with records $\mathcal{E}_j$ and a fixed higher-is-better score S , define

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

The two models are scored on the same records. This measured quantity is not population risk. For accuracy it is a sample average of paired correctness differences; for macro-F1 it is a difference of dataset-level scores, not an average of per-example losses. Metric scales are reported separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics). ADAPT commits f_a when the interval supports positive benefit.

FREEZE rejects the candidate when it supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection. Abstention is an unresolved assessment. An operator can attach monitoring, label acquisition, or human review to the abstention record; the experiments here evaluate only the frozen fallback and uncertainty log, not those additional interventions. Retaining f_0 after missing evidence must be logged as ABSTAIN, not as a certified FREEZE.

The observed-cell oracle chooses the larger score. Regret is the policy’s score shortfall from that oracle, averaged over the named units. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA uses the candidate score only after ADAPT. Lower regret is better, and a zero-benefit tie carries zero regret for either model even though it supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

These are three different measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\text{ADAPT}, \text{FREEZE}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon), \quad (3)$$

$$\text{FA}_u = \mathbb{P}(g = \text{ADAPT}, \Delta^{\text{cell}} \leq 0). \quad (4)$$

Conditional false adaptation is $\text{FA}_c = \mathbb{P}(\Delta^{\text{cell}} \leq 0 \mid g = \text{ADAPT})$, defined only when $\mathbb{P}(g = \text{ADAPT}) > 0$. The operational proposition below bounds the marginal error under interval coverage; it does not bound FA_c . A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. Analogously, a false FREEZE is a FREEZE decision with $\Delta^{\text{cell}} \geq 0$; its conditional frequency divides by FREEZE decisions and is undefined when that action is absent.

The population theorem uses binary labels and 0/1 loss. The empirical gate can instead target any predeclared scalar score difference, including multi-class metrics. This interface does not extend the binary theorem. In general multiclass classification, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, with $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$; only the binary setting forces $p_0 = 1 - p_a$.

Table 1: Closest roles, targets, and remaining distinctions. Literature comparisons are conceptual here; a historical AETTA comparison is reported in Appendix B.3.4, as a descriptive comparison whose rank-clipped calibration does not establish nominal coverage, while native TTA/ALine and full POEM comparisons remain incomplete.

Approach	Main target	Relation to this paper
TTA/ALine; AETTA	adapted-model performance and selection	closest empirical estimators; their scores can inform paired comparisons
POEM; risk monitors	protected adaptation or running risk	different monitored target and sequential assumptions
Partial-identification decisions	uncertain payoffs across compatible laws	context for the population benefit set and robust sign decision
Selection/shift-conditional conformal	coverage after selection or within a shift class	stronger, specialized targets not supplied by KGA’s marginal interval
K-Bound / KGA	feasible population benefit / predicted cell benefit	shared update question, distinct uncertainty objects and guarantees

Table 2: Notation and probability targets. Population and measured-cell quantities are kept separate.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D, \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
γ, β	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\hat{\Delta}, \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

4 Population Partial-Identification Frontier

The result describes which benefit signs are compatible with a fixed label-free evidence law and a declared bound on the hidden residual. The population analysis uses binary classification and 0/1 loss, where disagreement implies that exactly one predictor is correct. This permits the exact benefit reduction and strict-commitment frontier derived below. The impossibility result applies within the declared label-kernel class; it does not rule out certification under additional structural restrictions or establish the same frontier for general multiclass problems. KGA separately evaluates an empirical scalar-benefit gate, including on multiclass benchmarks.

4.1 Disagreement-Region Model

Assumption 1 (Deployment setup). Let $Y \in \{0, 1\}$, use 0/1 loss, fix measurable binary predictors f_0, f_a , and assume $\mu_T(D) > 0$ for $D = \{x : f_0(x) \neq f_a(x)\}$. Let $s : \mathcal{X} \rightarrow [0, 1]$ be a fixed measurable candidate-correctness score, fitted using source information, and

let the target law admit the measurable pointwise correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. On D , set

$$M = \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \quad \gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]. \quad (5)$$

The declared population class is $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$ for an externally supplied $\beta \geq 0$. Because $s \in [0, 1]$, the feasible margin satisfies $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities and frontier below do not require a source-calibration property of s .

Remark 1 (Residual identity). The definition gives

$$\begin{aligned} M + \gamma &= (\mathbb{E}[s(X) \mid D] - \frac{1}{2}) + \mathbb{E}[\eta_a(X) - s(X) \mid D] \\ &= \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}. \end{aligned}$$

This is an identity, not empirical evidence that calibration transfers. We call γ a target disagreement-conditional calibration residual, not automatically source-to-target drift. Ordinary source calibration need not hold after conditioning on D , so γ can be nonzero even when $P_T = P_S$. A literal label-kernel shift interpretation would require an additional reference, for example the pointwise source correctness

Table 3: Three claim levels and their validity obligations. This table collects the distinctions used throughout the paper.

Claim level	Required premise	Conclusion and boundary
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class, not a learned residual bound
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target, not automatic conditional or anytime control
Empirical evaluation	fixed recorded candidates, splits, metric and action rule	measured cell regret, interval inclusion and action exposure; population transfer needs a separate argument

$s(x) = \mathbb{P}_S(f_a(X) = Y \mid X = x)$. The no-shift counterexample is given in Appendix L. Calibration notions also depend on the conditioning information: top-label calibration, for example, additionally conditions on the predicted class [43]. Neither score nor top-label calibration automatically supplies calibration conditional on D .

Definition 2 (Risk alignment). Evidence Z is risk-aligned on a class $\mathcal{P}$ if no two laws in $\mathcal{P}$ with opposite nonzero benefit signs induce the same law of Z . This is weaker than uniform strict-sign identification: a fibre containing zero and positive benefit is risk-aligned by this definition, but does not support a uniformly positive claim. Population frontier claims take M to be fixed on each augmented evidence fibre, equivalently by using $\tilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is uniformly directionally sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$; FREEZE is sound when $\Delta(P) < 0$ for every such law. A pointwise maximal three-way rule commits whenever one strict action is sound and abstains otherwise. The strict convention treats $\Delta = 0$ as insufficient for either directional claim, although the actions then have equal task risk.

4.2 Label-Kernel Freedom on the Disagreement Region

Lemma 1 (Label-kernel freedom on disagreement). Fix $(\mu_T, P_S, f_0, f_a, s)$, hence D and M , and fix the conditional label law outside D . For every measurable $\eta : D \rightarrow [0, 1]$ there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D . Every label-free observable formed from the input batch, the fixed models, and their outputs has the same law for all such η , while

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

Proof. On D , binary predictors disagree, so their outputs are 0 and 1. Assign conditional mass $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$, retaining the fixed kernel outside D . This changes no input marginal or model output and therefore changes no label-free evidence law. The expression for γ follows from Eq. (5). $\square$

4.3 Benefit Reduction and Matched-Evidence Impossibility

Lemma 2 (Disagreement-region reduction). Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a = Y \mid D)$,

$$\begin{aligned} \Delta &= 2\mu_T(D)(\bar{a} - \tfrac{1}{2}), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \tfrac{1}{2}) = \text{sign}(M + \gamma). \end{aligned}$$

The identity $\text{sign } \Delta = \text{sign}(M + \gamma)$ holds for every target law; it does not assume $|\gamma| \leq \beta$.

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ ,

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}, \end{aligned}$$

so $\text{sign } \Delta = \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma)$. $\square$

Theorem 1 (Interior matched-evidence impossibility). Fix $\beta > 0$. For any feasible margin $M \in [-\frac{1}{2}, \frac{1}{2}]$ with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that $\text{Law}(\tilde{Z} \mid P_T^1) = \text{Law}(\tilde{Z} \mid P_T^2)$ for $\tilde{Z} = (Z, M)$, $M(P_T^1) = M(P_T^2) = M$, and $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\text{sign } \Delta = \text{sign } M$ is determined and no such pair exists: matched-evidence ambiguity requires a positive declared residual bound.)

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$,

holding the off- D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ , and: *class membership*, $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \leq \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence*, $\text{Law}(W \mid P_T^\eta)$ does not depend on η by the lemma and $\tilde{Z} = (Z, M)$ is a function of W , so $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$; *opposite signs*, Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\theta\delta \neq 0$ with $\text{sign } \Delta_\theta = \theta$. $\square$

The construction changes only the target label kernel on D , leaving the label-free evidence laws exactly identical. The result is therefore an exact non-identifiability statement rather than an estimation-rate bound.

Corollary 1 (Matched-evidence abstention lower bound). *Fix $\alpha \in (0, \frac{1}{2})$. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule g satisfying $\mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] \leq \alpha$ in every admissible world and $\mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] \leq \alpha$ in every admissible world must satisfy $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. $\square$

4.4 Boundary Case

Proposition 1 (Boundary case and closed-band abstention). *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band $|M| \leq \beta$, and any label-free rule controlling both marginal errors at level α abstains there with probability at least $1 - 2\alpha$. Opposite nonzero signs are asserted only in the interior $|M| < \beta$.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so both directional error events $\{\text{ADAPT}, \Delta \leq 0\}$ and $\{\text{FREEZE}, \Delta \geq 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\text{ADAPT}] \leq \alpha$ and $\mathbb{P}[\text{FREEZE}] \leq \alpha$, hence $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class

$\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. $\square$

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

Theorem 2 (Exact strict-commitment frontier). *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \geq 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

- (i) **Reduction.** *For every P_T , $\text{sign } \Delta = \text{sign}(M + \gamma)$ (Lemma 2).*
- (ii) **Sufficiency.** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then $\text{sign } \Delta = \text{sign } M$.*
- (iii) **Necessity.** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite nonzero benefit signs; at $|M| = \beta > 0$ the admissible residual $\gamma = -\text{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no strict benefit direction is uniformly supported by $\text{Law}(\tilde{Z})$ over $\mathcal{C}_\beta$ whenever $|M| \leq \beta$.*
- (iv) **Strict-commitment iff.** *At the fixed augmented-evidence law, a strict action in $\{\text{ADAPT}, \text{FREEZE}\}$ is uniformly directionally sound in the sense of Definition 3 if and only if $|M| > \beta$; on the closed band $|M| \leq \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the pointwise maximal sound action. The population rule $M > \beta \Rightarrow \text{ADAPT}$, $M < -\beta \Rightarrow \text{FREEZE}$, $|M| \leq \beta \Rightarrow \text{ABSTAIN}$ is the pointwise maximal sound certificate on $\mathcal{C}_\beta$.*

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \leq \beta$, then $M + \gamma > 0$, so $\text{sign } \Delta = \text{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\text{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. $\square$

The feasible range can also be read directly from the identified benefit set. Write $d = \mu_T(D) > 0$. Over the full declared class with the observables fixed,

$$\Delta(\mathcal{C}_\beta) = 2d \left[\max\{-\tfrac{1}{2}, M - \beta\}, \min\{\tfrac{1}{2}, M + \beta\} \right].$$

Indeed, $\bar{a} - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ and $|\bar{a} - \frac{1}{2} - M| \leq \beta$; every value in their intersection is attained by a constant correctness kernel on D . A strict direction is available exactly when this whole interval lies above or below zero.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$ M < \beta$	both strict signs are feasible	ABSTAIN
$ M = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem depends on a declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual; it is not a conservative default. Without a credible upper bound on $|\gamma|$, the population frontier cannot support a robust strict action. More unlabeled samples do not solve this problem because the missing difference lies in the target labels, not in the input batch.

Empirical abstention is not automatically evidence of structural unknowability. A wide finite-sample interval may instead reflect limited calibration data, estimator error, or transfer failure. The frontier therefore leaves one central question: can its required residual bound be learned from the same label-free evidence?

5 What Label-Free Evidence Can-not Learn

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, the fixed models and score function, label-free statistics derived from them, and any source or calibration sample whose law is specified independently of the target label kernel. An audit is a nonnegative random quantity $\hat{\beta} = g(W, U)$, where U is independent randomization.

For a declared class $\mathcal{C}$ and an observable law $z = \text{Law}(W)$, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}(W \mid P) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (6)$$

The fibre collects target laws that the audit cannot distinguish. Its radius is therefore the smallest uniform residual bound that those observations could support.

5.2 Exact Fibre-Wise Audit Floor

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ and $\delta \in [0, 1]$. If $\hat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$, then*

$$\mathbb{P}(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the high-probability event above, an audited frontier commits only where the frontier using the declared radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. Every law in the fibre induces the same law for W , and hence the same law for $\hat{\beta}$. Validity for each fixed P requires that common law to exceed $|\gamma(P)|$ with probability at least $1 - \delta$. Taking an increasing sequence of attainable absolute residuals approaching the supremum gives the lower bound. The constant audit is valid by the definition of the supremum. If $\hat{\beta} \geq \Gamma_z$, then $|M| > \hat{\beta}$ implies $|M| > \Gamma_z$. $\square$

This result does not say that useful budgets never exist. It says that the same label-free fibre cannot create information about the target label kernel that the fibre definition removes. A valid audit must cover the unresolved radius; reading more functions of the same W does not shrink it. The constant $\Gamma_z(\mathcal{C})$ is an oracle benchmark for the fixed evidence-law fibre. The theorem does not provide an algorithm that learns the unknown law z or its radius from a finite observed batch. We therefore describe this as a fibre-wise audit floor rather than an unspecified minimax estimation result.

5.3 Declared Budget and Abstention Radius

For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (7)$$

Thus for any $\beta \in [0, 1/2]$, the least fibre-wide budget a label-free audit can certify is the budget already

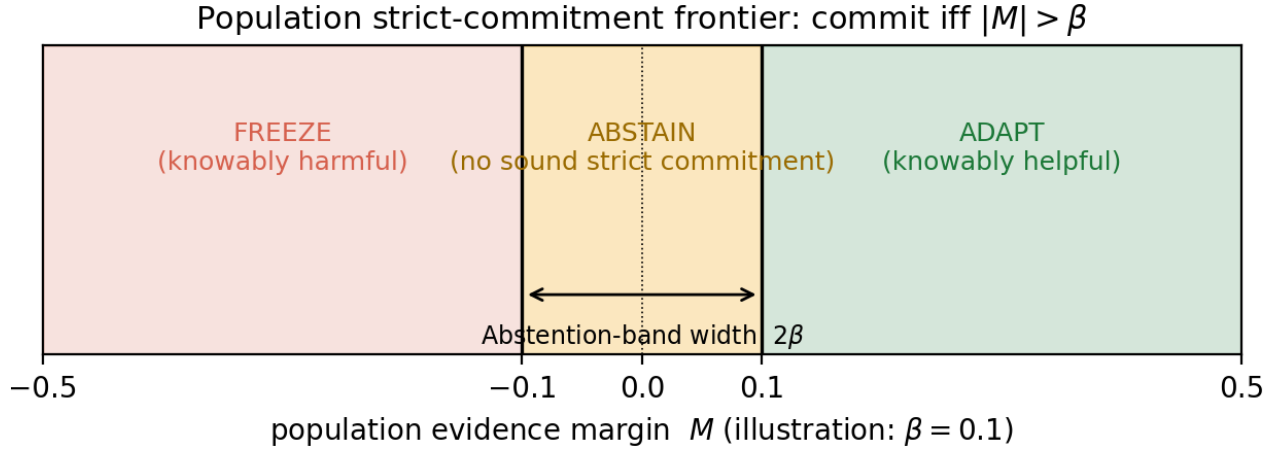


Figure 2: Population strict-commitment frontier. $M > \beta$ supports adapt; $M < -\beta$ supports freeze; and $|M| \leq \beta$ supports no uniform strict commitment. Interior ambiguity uses opposite nonzero benefit signs, whereas the boundary uses zero-benefit ambiguity. This population rule is not the numerical KGA rule used on the real-data panels.

declared by the class. The same number is also the population abstention radius:

$$\begin{aligned} & \text{least valid fibre-wide residual bound} \\ & = \text{population abstention radius} = \beta \quad (0 \leq \beta \leq \tfrac{1}{2}) \end{aligned}$$

This equality connects the audit problem to the strict-commitment frontier. For the full declared class, Theorem 3 rules out a uniformly tighter residual budget from the same evidence. We therefore treat β as an *externally justified sensitivity parameter*, by analogy to Rosenbaum’s sensitivity parameter Γ for unobserved confounding in observational studies [55] or the ambiguity set radius in distributionally robust optimization (an analogy, not a mathematical equivalence). Extended exploratory formulations of break-even sensitivity frontiers and continuous selection odds-ratio curves are preserved in the supplementary research archive; the primary manuscript retains the exact population frontier $\Delta(\mathcal{C}_\beta)$ established in Theorem 2.

5.4 Historical Labels Require a Transfer Assumption

Suppose the audit also observes a fully labeled calibration domain, but the declared class contains no coupling between that domain’s label kernel and the deployment label kernel. The construction in Lemma 1 changes only deployment labels on D and leaves the calibration sample unchanged. The augmented evidence law is therefore identical across the same target fibre, so Theorem 3 applies without a smaller radius.

Labels help only when they come from the deployment domain or when a stated assumption links the labeled and deployment domains.

5.5 What Additional Structure Can Buy

The missing information can be supplied, but each route changes the problem. Deployment labels can estimate target correctness directly. Historical labeled deployments can support an episode-marginal budget under an exchangeability model. A structural coupling can bound the change between calibration and target domains. Domain knowledge can declare a physically meaningful limit.

Multiclass extensions under entropy-constrained support conditions require separate assumptions and proofs; they are preserved as unproved exploratory research notes outside the maintained claims of this manuscript.

5.6 Implications for K-Bound and KGA

The evidence-fibre result motivates a practical question: what additional information can support an update decision? KGA uses labeled historical shifts to predict and calibrate measured benefit. Section 6.1 states its inferential target and the extra assumptions required for population interpretation.

Table 4: Correspondence between population theory (K-Bound) and the empirical decision rule (KGA) addressing the same paired deployment choice. Distinct inferential targets do not make the quantities equal.

Quantity	Role in Population Frontier (K-Bound)	Status in Empirical Rule (KGA)
M	Observable margin on disagreement region	Not explicitly estimated
γ	Hidden calibration residual ($M + \gamma = \bar{a} - \frac{1}{2}$)	Unobserved / unestimated
β	Externally declared hidden-residual bound	Not a numerical KGA input
Δ	Population benefit $2\mu_T(D)(M + \gamma)$	Target estimand under population transfer
Δ^{cell}	Not present in pure population model	Measured evaluation-cell benefit $S(f_a; \mathcal{E}) - S(f_0; \mathcal{E})$
$\hat{\Delta}$	Not present in pure population model	Fitted prediction $h_\theta(Z)$ of empirical target Δ^{cell}
ε	Not present in pure population model	Empirical residual-calibration radius
$b(m, \delta)$	Not present in pure population model	Separately justified sampling allowance

6 KGA: An Empirical Benefit-Interval Gate

6.1 From Population Identification to Empirical Decisions

K-Bound characterizes population benefit over a declared class of target laws. KGA predicts measured evaluation-cell benefit using labeled historical outcomes and a separately calibrated residual interval. It does not numerically apply $|M| > \beta$, and its empirical radius ε is not an estimate of the population residual bound β . When a credible population budget is supplied externally, a direct finite-sample frontier rule is available (Appendix A.6). The empirical route instead depends on support for its calibration assumptions.

The binary population model gives

$$\Delta = 2\mu_T(D)(M + \gamma), \quad (8)$$

while the observed cell outcome and fitted empirical prediction are

$$\Delta^{\text{cell}} = S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \quad \hat{\Delta} = h_\theta(Z). \quad (9)$$

Table 4 summarizes this architectural correspondence.

A population interpretation requires both cell-level coverage and a separate sampling bound:

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b(m, \delta)) &\geq 1 - \delta. \end{aligned}$$

The cell-coverage and fresh-sample concentration

premises imply, by a union bound,

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b(m, \delta)) \\ \geq 1 - \alpha_{\text{cell}} - \delta. \end{aligned}$$

The enlarged interval therefore covers population benefit only when both premises hold. Proposition 3 states this implication; its sampling assumptions are not established for the benchmark panels. Table 4 keeps the population and empirical quantities distinct.

KGA is label-free at deployment, not during development and calibration. Historical labeled cell outcomes fit h_θ and supply residuals. At the new decision, KGA reads only Z and the frozen fitted artifacts. The interval premise for the empirical target is

$$\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha. \quad (10)$$

The static interval requires its stated calibration assumptions. Sequential monitoring is a separate extension requiring its own null model and validity conditions. The empirical results reported here do not establish an anytime guarantee or a guaranteed response to every harmful shift.

6.2 The Coverage-to-Action Proposition

The following standard implication makes the decision semantics explicit. It is not a new calibration theorem: coverage is the premise, and establishing it is the statistical burden.

Proposition 2 (Coverage-to-action implication). *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\hat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\hat{B}$ and radius ε must be available without reading the new unit's evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{B} - B| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g = \begin{cases} \text{ADAPT} & \hat{B} - \varepsilon > 0, \\ \text{FREEZE} & \hat{B} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned} \mathbb{P}[g = \text{ADAPT}, B \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, B \geq 0] &\leq \alpha. \end{aligned}$$

Proof. On the event $\{|\hat{B} - B| \leq \varepsilon\}$, the condition $\hat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α . The freeze case is symmetric. $\square$

Remark 2 (Unit scaling: empirical radius versus theoretical budget). The empirical radius ε and theoretical residual bound β live on fundamentally different scales. The population identity is $\Delta = 2d(M + \gamma)$ where $d = P_T(D) > 0$ is the disagreement probability and $2d$ is twice that probability. Thus, $|\Delta - 2dM| = 2d|\gamma| \leq 2d\beta$. The unclipped population envelope centered at $2dM$ has radius $2d\beta$ on the benefit scale, not β ; the exact population benefit set $\Delta(\mathcal{C}_\beta)$ in Theorem 2 includes clipping to $[-d, d]$: $\Delta(\mathcal{C}_\beta) = 2d[\max(-\frac{1}{2}, M - \beta), \min(\frac{1}{2}, M + \beta)]$. The quantities have different definitions. The declared bound β limits the absolute target disagreement-conditional residual, whereas ε is calibrated from prediction errors for the empirical target. They may coincide numerically, but that does not make them interchangeable.

Proposition 3 (Cell-to-population transfer via compound bound). *Let $m \geq 1$ be an integer and let $\Delta^{\text{cell}} = \frac{1}{m} \sum_{j=1}^m W_j$ be the measured cell benefit on m fresh, conditionally i.i.d. evaluation draws, where $W_j = \mathbf{1}\{f_a(X_j) = Y_j\} - \mathbf{1}\{f_0(X_j) = Y_j\} \in [-1, 1]$, with conditional mean Δ . Suppose an independently fitted benefit predictor h and split-conformal radius ε achieve marginal cell coverage $\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha_{\text{cell}}$ over exchangeable deployment episodes. For a sampling budget $\delta \in (0, 1)$, define $b(m, \delta) = \min\{2, \sqrt{2\log(2/\delta)/m}\}$. Then the compound interval $I_{\text{population}} = [\hat{\Delta} - (\varepsilon + b), \hat{\Delta} + (\varepsilon + b)]$ satisfies*

$$\mathbb{P}(\Delta \in I_{\text{population}}) \geq 1 - \alpha_{\text{cell}} - \delta.$$

The corresponding population decision rule g_{pop} using radius $\varepsilon + b$ satisfies

$$\mathbb{P}(\{g_{\text{pop}} = \text{ADAPT}, \Delta \leq 0\} \cup \{g_{\text{pop}} = \text{FREEZE}, \Delta \geq 0\}) \leq \alpha_{\text{cell}} + \delta.$$

Proof. By Hoeffding’s inequality for bounded variables in $[-1, 1]$ of range 2, $\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) \geq 1 - \delta$. On the intersection of cell coverage and concentration, the triangle inequality yields $|\hat{\Delta} - \Delta| \leq |\hat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \leq \varepsilon + b$. A union bound proves coverage at level $1 - \alpha_{\text{cell}} - \delta$. Event containment (as in Proposition 2) immediately bounds the probability of an incorrect strict direction by $\alpha_{\text{cell}} + \delta$. $\square$

Remark 3 (Direct empirical sampling of the population frontier). An alternative route directly operationalizes the population frontier $|M| > \beta$. Fix the predictor pair and score independently of a fresh i.i.d. sample of unlabeled target inputs, and let $n_D > 0$ of those inputs fall in D . The empirical disagreement margin $\hat{M} = \frac{1}{n_D} \sum_{X_i \in D} s(X_i) - \frac{1}{2}$ concentrates with radius $r_M = \sqrt{\log(2/\alpha)/(2n_D)}$. Under the externally declared bound $|\gamma| \leq \beta$, the rule $\hat{M} - r_M > \beta \Rightarrow \text{ADAPT}$, $\hat{M} + r_M < -\beta \Rightarrow \text{FREEZE}$, and ABSTAIN otherwise, guarantees directional error at most α . This couples sampling uncertainty r_M additively with structural uncertainty β .

Remark 4 (Marginal vs. conditional error). Proposition 2 bounds the *marginal, unconditional* false-commit probabilities for the declared target B . Its false-adapt rate is $\text{FA}_u = \mathbb{P}[g = \text{ADAPT}, B \leq 0]$. It does *not* bound $\text{FA}_c = \mathbb{P}[B \leq 0 \mid g = \text{ADAPT}]$, defined when $\mathbb{P}[g = \text{ADAPT}] > 0$ and reported descriptively for $B = \Delta^{\text{cell}}$. Selection-conditional coverage requires a separate construction, such as that of Jin and Ren [52]; conditional guarantees over specified shift classes also require their own calibration procedure [53]. Neither is supplied by the marginal premise above. Simultaneous protection across candidates and time-uniform protection likewise require additional guarantees. Risk alignment (Definition 2) concerns population sign identifiability; coverage for B alone yields the *interval* implication above.

Remark 5 (Fallback when assumptions are unsupported). If interval coverage for the declared target and evaluation unit—or the assumptions supporting it—cannot be justified, Proposition 2 supplies no level- α protection for either strict action. A separately declared operational fallback may retain the frozen model and record abstention. Serving the frozen model without a valid negative interval is not a certified FREEZE decision. This policy recommendation does not establish that the retained model is safe or optimal. If *risk alignment* (Definition 2) cannot be justified for an intended population sign claim, that interpretation remains unsupported. The interval implication of Proposition 2 still holds when coverage for the declared B is established (cf. Theorem 1).

Proposition 3 formalizes the conditional bridge to population risk Δ : under independent fresh evaluation draws, adding the Hoeffding concentration radius $b(m, \delta)$ to the empirical radius ε yields a compound population interval with radius $\varepsilon + b$ and error budget $\alpha_{\text{cell}} + \delta$. The full protocol, counterexample, and proof are detailed in Appendix A. No such sampling-

Table 5: 108-cell ImageNet-C population replay: cell gate ($\hat{\Delta} \pm \varepsilon$) vs. compound population interval ($\hat{\Delta} \pm (\varepsilon + b)$) under 5-fold cross-fitting ($\alpha_{\text{cell}} = \delta = 0.05$, $\alpha_{\text{pop}} = 0.10$, $m = 2,000$, $b = 0.0607$). Full 12-column breakdown with fold details and radii ranges in Table 16.

Candidate	N	Regret (Cell / Pop)	Cell Gate (A / F / U)	Compound Pop (A / F / U)	FA_u
Tent	36	0.0162 / 0.0468	18 / 0 / 18	1 / 0 / 35	0.000
EATA	36	0.0095 / 0.0386	20 / 0 / 16	0 / 0 / 36	0.000
SAR	36	0.0245 / 0.0267	2 / 11 / 23	0 / 10 / 26	0.000
Total / Panel	108	0.0167 / 0.0374	40 / 11 / 57	1 / 10 / 97	0.000

error bound is established for the reported panels; in particular, when a candidate adapts to evaluation inputs, transductive normalization couples the candidate across batches.

The formalized action rule proves that a strict decision has the correct sign whenever the compound interval contains population benefit. This implication does not establish the sampling or calibration premises for the benchmarks; the scope and theorem mapping are given in Appendix M.

ImageNet-C 108-cell population replay. To audit this procedure on existing benchmark data without changing historical estimands, we replayed the outcome-blind compound interval $[\hat{\Delta} \pm (\varepsilon + b)]$ across 108 ImageNet-C development cells (36 cells each for Tent, EATA, and SAR; five-fold cell-outcome-disjoint cross-fitting; sensitivity parameter $m = 2,000$, yielding uniform concentration radius $b = 0.0607$; Table 5). Full fold breakdowns and radius distributions appear in Table 16 (Appendix A.7). Finite radii were obtained on 108/108 cells. Across the panel, the cell-level gate ($\hat{\Delta} \pm \varepsilon$) issued 40 ADAPT, 11 FREEZE, and 57 ABSTAIN decisions with zero false adaptations ($\text{FA}_u = 0.0000$) and mean oracle regret 0.0167. Incorporating the sampling radius ($b = 0.0607$, with mean compound half-width $\bar{\varepsilon} + b = 0.1272$) shifted compound decisions to 1 ADAPT, 10 FREEZE, and 97 ABSTAIN, preserving zero observed false adaptations ($\text{FA}_u = 0.0000$) while shifting regret to 0.0374. This concretely demonstrates the sensitivity to the added concentration radius $b = 0.0607$: because most empirical gains on this benchmark are smaller than the universal Hoeffding penalty for $m = 2,000$, the compound rule withholds action on nearly all adaptations, illustrating the additional abstention induced by this sampling allowance rather than establishing prospective population truth. As clarified in Appendix A.7, $m = 2,000$ is an evaluation sensitivity parameter, not fresh sampling; population truth remains unestablished on historical development records.

Non-oracle population diagnostic. While the synthetic diagnostic in Table 15 tested exact Bayes prediction ($\Delta(Z) = 0.6Z$), evaluating an imperfect predictor subject to estimation noise requires an extended non-oracle study. We conducted a diagnostic across 1,000 independent trials per configuration over 16 (N_{cal}, m) pairs, fitting an imperfect linear predictor on finite development episodes ($m = 32$; Table 14, Appendix A.7). An unhedged point rule ($\text{sign}(\hat{\Delta})$) incurs a 30.9%–35.6% directional error rate ($\{g = \text{ADAPT} \wedge \Delta \leq 0\} \cup \{g = \text{FREEZE} \wedge \Delta \geq 0\}$) with 0% abstention (where FA_u is defined strictly for false ADAPT; here directional errors combine false ADAPT and false FREEZE on null/tie states); crucially, 100% of these errors occur on null/tie states ($Z = 0, \Delta = 0$) due to estimation noise around zero, with zero errors on strictly harmful states ($\Delta = -0.6$). In the finite-radius configurations, the calibrated cell interval recorded 93.9%–96.5% empirical cell inclusion against a nominal 95% level and zero observed wrong-direction decisions. Observed population inclusion was approximately 100% for the cell interval in this synthetic model. The compound interval also recorded 100% empirical population inclusion and zero observed wrong-direction decisions. Its abstention rate was 99.9%–100% at $m = 32$ and approximately one third for the finite-radius configurations with $m \geq 128$. At $N_{\text{cal}} = 18$, the required calibration rank is unavailable, so the infinite-radius rule abstains throughout. These are finite-simulation observations, not a proof of universal coverage.

6.3 Evidence, Calibration, and the Action Rule

The controlled implementation stores 11 label-free features, with algebraic rank at most 9 because two features are deterministic differences. They summarize confidence, class balance, model disagreement, and update size. The current principal CIFAR replay fits a separate `GradientBoostingRegressor` [33] for each adapter, stream seed and score fold, using squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and model seed 0. The later shared-predictor comparison has a separately specified estimator configuration (Appendix N.1). Natural protocols retain their own fitted model and feature schema; for example, CCT-20 uses a ridge model. Details are in Appendix I. A common interface does not imply that an estimator or radius transfers across adapters.

Given residuals $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \leq$

$\dots \leq r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (11)$$

Split-conformal coverage requires joint exchangeability of the calibration and next-unit scores under the fitting and candidate-construction protocol [36, 37]. An independently fitted predictor and exchangeable calibration/deployment units are one sufficient route. The rank counts evaluation units, not the images inside them.

The paper distinguishes current outcome-disjoint calibration, Protocol B, from the historical leave-one-cell-out replay, Protocol A.

Protocol B: Three-way cell-outcome-disjoint cross-fitting (used in the current primary CIFAR table, interval diagnostics, the 108-cell ImageNet-C population replay, and the Appendix B.3 27-cell audit). For each score fold, its entire complement is partitioned into strictly disjoint estimator-fit and residual-calibration subsets, ensuring that scored cell outcomes enter neither predictor training nor conformal calibration. Historical saved LOO predictions and radii do not determine these current CIFAR results.

Protocol A: Leave-one-cell-out empirical order-statistic calibration (used in historical controlled grids). Benefit predictions $\hat{\Delta}_i$ were cross-fitted on the remaining $n-1$ cells, and each scored residual $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$ was excluded from the pool used to compute its own radius ε_i . However, this historical leave-one-out construction is peer-dependent: for $j \neq i$, the model h_{-j} was fitted on training data containing Δ_i^{cell} , so the scored cell’s outcome can indirectly influence other residuals r_j in its calibration pool through model fitting. Where historical stress grids retain pooled LOO residuals (Protocol A), they are reported with this peer-dependence caveat rather than claiming full outcome disjointness. Natural protocols use separated development, calibration, and target partitions where the archived design records that separation.

For example, prediction 0.04 and radius 0.01 give [0.03, 0.05] and ADAPT; prediction -0.04 gives $[-0.05, -0.03]$ and FREEZE; prediction 0.005 gives $[-0.005, 0.015]$ and ABSTAIN. These illustrate the rule, not measured results.

Candidate adaptation, frozen and candidate inference, evidence extraction, and shadow-state storage all contribute to runtime. The tree prediction and interval comparison are inexpensive, but the whole procedure is not one extra forward pass. Historical synchronized timing and peak-memory evidence

Algorithm 1 KGA for one proposed update

Require: Frozen f_0 , adapter $\mathcal{A}$, unlabeled X , frozen h_θ and calibration artifacts

- 1: Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
 - 2: Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
 - 3: **if** the assessment is unavailable **then**
 - 4: retain f_0 , log the reason, and return ABSTAIN
 - 5: **end if**
 - 6: Obtain ε from the calibrated residual rule in Eq. (11)
 - 7: $\hat{\Delta} \leftarrow h_\theta(Z)$; $L \leftarrow \hat{\Delta} - \varepsilon$; $U \leftarrow \hat{\Delta} + \varepsilon$
 - 8: **if** $L > 0$ **then**
 - 9: commit f_a and return ADAPT
 - 10: **else if** $U < 0$ **then**
 - 11: discard f_a and return FREEZE
 - 12: **else**
 - 13: retain f_0 , log uncertainty, and return ABSTAIN
 - 14: **end if**
-

was not recovered; the retained literals are therefore audit records rather than performance claims (Table 45). Workload-specific measurements reported in the supplement have a distinct scope, and comparable end-to-end deployment latencies across the historical benchmark tracks remain unestablished.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The controlled study asks whether the gate distinguishes useful from harmful updates. The CCT-20 study asks whether a frozen gate avoids measured candidate harm under a natural camera-location shift. These are different claims. We predefine the interpretation of each study before presenting its results in Table 6.

CIFAR-10-C [23] supplies 2,160 recorded cells per candidate: five archived ResNet-18 run-seed groups, each evaluated across 432 corruption/composition/update conditions. The current gate is refitted from the saved label-free features Z and measured cell outcomes under Protocol B, rather than reusing historical stored predictions or radii. For this canonical evaluation, the analysis uses three-way cell-outcome-disjoint cross-fitting: each scored fold has separate estimator-fit and residual-calibration subsets, so its outcome enters neither its fitted prediction nor its radius. This is still a retrospective opened

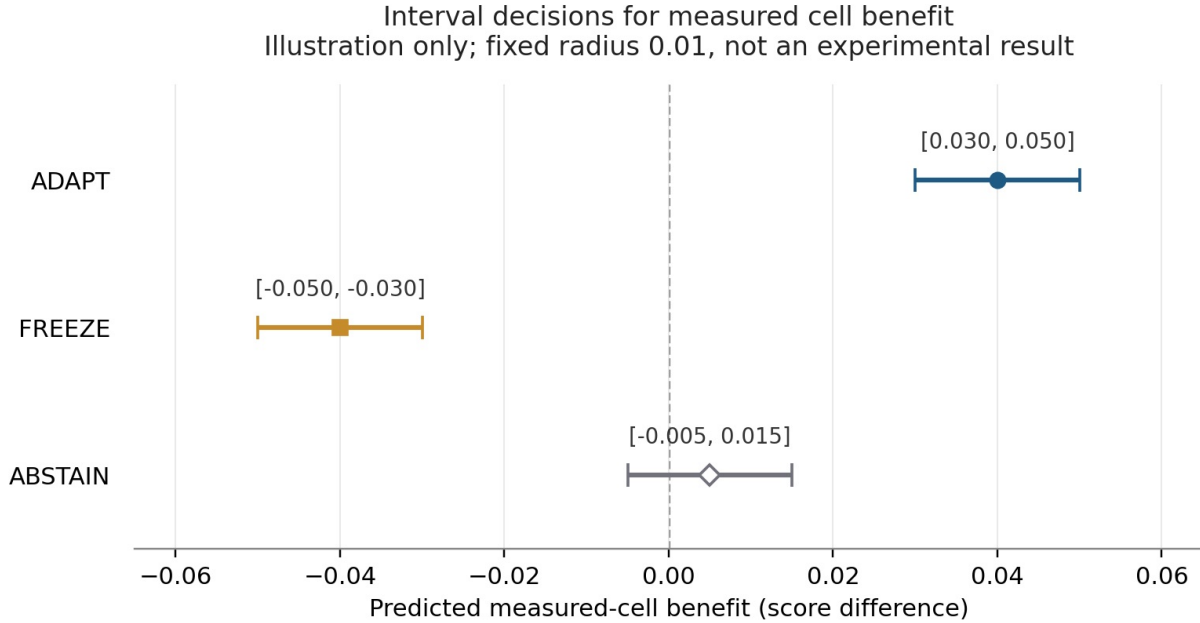


Figure 3: Illustration of the three interval decisions for measured cell benefit. Only an interval strictly above or below zero supports a commitment. The figure illustrates the rule; coverage must be established for the intended evaluation unit.

Table 6: Experiment status and role. Primary presentation does not imply prospective confirmation. All recorded unfavorable outcomes remain disclosed.

Evidence	Design and status	Question it can answer
CIFAR-10-C: Tent, EATA, SAR	opened controlled grids; cross-fitted empirical calibration	candidate-specific routing on the evaluated cells
CCT-20: Tent	internally sealed protocol; five checkpoints and nine target camera locations	conservative retention under the stated score and predeclared utility-retention rule
So2Sat-LCZ42 v1/v2	v1 development stop; v2 calibration-screen failure; targets unopened	whether the respective pre-target eligibility checks were met
Other replayable tracks	opened natural/semantic and candidate-configuration diagnostics	transfer limits and one-sided retention; full results in the supplement
PACS; iWildCam; historical ports; focused Office-Home route	partial replay, withheld metric, supervised policy, or invalid route	audit history only where the corresponding contract is incomplete or invalid

panel, not an untouched environment test; historical stored fields remain audit-only and do not determine the current decisions.

7.2 CCT-20 Design and Utility Retention

The CCT-20 camera-location study [48] crosses 5 independent source checkpoints with 9 target locations, giving 45 checkpoint–location cells. It uses 6950 probe images and 16325 evaluation images (23275 total). The primary metric is set-membership top-1 accuracy: a prediction is correct if it matches any distinct archived category for that image. It is not an official single-label leaderboard reproduction. The candidate uses the official Tent implementation, one

non-episodic Adam step at learning rate 10^{-3} , and state reset per cell.

The protocol was internally sealed before source training and outcome access, not publicly registered. Aggregate target label metadata had been inspected during candidate ranking; cell-level outcomes remained unopened until one-shot scoring. This is outcome-blind execution, not a label-unopened study. The source ridge fit used 10 rows from two camera locations; calibration used 45 rows aggregated to nine cis-test locations. Transfer to trans-test locations remains an assumption: the protocol treats camera-location clusters as the population unit and assumes the nine trans-test clusters are exchangeable with the

declared target population under the locked sampling rule. This is a design assumption, not an empirically proven fact about future deployments.

For either fixed policy b , define G_b as mean baseline regret minus KGA regret. Positive G_b means lower regret for KGA. The *utility-retention* rule is a predeclared, limited harm-avoidance check. Using nominal pointwise 95% bootstrap lower endpoints, it passes when

$$L_{\text{adapt}} > 0, \quad L_{\text{freeze}} > -0.005. \quad (12)$$

The second threshold is the frozen-policy noninferiority margin, on the accuracy-proportion scale. This rule can pass when KGA always retains f_0 . The stronger routing criterion additionally requires improvement over both fixed policies at the declared inference level and nontrivial ADAPT/FREEZE exposure. This endpoint evaluates utility retention under the declared protocol; it is not a population-safety guarantee.

7.3 Scoring, Dependence, and Statistical Inference

All policies use the same evaluation records, and evaluation labels are joined only after actions are fixed. The candidate TTA step can be transductive: episodic candidates update on the unlabeled evaluation inputs, and online candidates may use evaluation-batch BatchNorm statistics. Development/calibration separation therefore does not imply independent fresh-sample inference.

We report oracle regret, action counts, commitment rate, and false-commit frequencies. Empirical interval inclusion and interval width are reported separately from commitment. A paired regret contrast always means baseline regret minus KGA regret, so positive values favor KGA. Accuracy and macro-F1 regrets are separated rather than pooled.

Bootstrap levels are nominal throughout. CIFAR sensitivity resamples six corruption families, conditional on an archived checkpoint. CCT-20 uses a paired two-way product bootstrap over checkpoint rows and location columns; sign-flip calculations instead use nine location means. For a vector G of cluster gaps, the sign-flip reference model is

$$G \stackrel{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \quad (13)$$

Independent zero-symmetric cluster gaps suffice. Exchangeability, marginal symmetry, or a single global sign symmetry alone is not sufficient [38]. Enumeration removes Monte Carlo error but does not verify

this null model. Retrospective Holm adjustment over the six prospectively named contrasts is reported, but Holm adjustment requires valid input p -values; nominal family intervals also require valid marginal coverage of their constituent intervals. The full inference specification and retrospective search status are collected in Appendix F.

8 Primary Results

8.1 Controlled Routing Depends on the Candidate

Table 7 reports all three CIFAR-10-C candidates under the same current rule. Tent and EATA have lower measured point regret than always-adapt and always-freeze. SAR is the counterexample: always-adapt has lower regret than KGA. The mixed panel therefore supports candidate-specific routing, not a claim that gating improves every adapter.

Tent regret is 0.0018, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA’s values are 0.0016, 0.0033, and 0.1313. In the canonical primary panel (Table 7), their actions are 1,094/334/732 and 1,207/118/835, respectively; both have zero observed false adaptations ($\text{FA}_u = 0.0000$). SAR commits 1,430/0/730 actions with regret 0.0017, versus 0.0003 for always-adapt ($\text{FA}_u = 0.0005$, 1 false adaptation).

On the recorded CIFAR SAR panel, always-adapt has very low paired-oracle regret. KGA’s higher regret reflects the net benefit forgone on cells where it retains the frozen predictor. The current results do not isolate whether those decisions arise mainly from prediction error, conservative calibration, or candidate-update dynamics. Normalization and sample-filtering effects remain hypotheses rather than established causes.

In contrast, Tent and EATA suffer degradation under severe corruptions, allowing KGA’s selective freezing and abstention to achieve net regret reductions over always-adapt (+0.0062 for Tent, +0.0017 for EATA on the canonical primary panel; matching the synchronized baseline study). Figure 4 illustrates the effect of changing helpful and harmful mixture weights in a preserved historical Tent seed-0 run. Its constructed mixture curve is separate from the current Protocol B aggregate above; it is not an additional deployment or a new cross-fitting evaluation.

A retrospective cluster-bootstrap sensitivity analysis across the six observed CIFAR-10-C corruption families (Table 34) indicates an estimated paired regret reduction for Tent of +0.0062

Table 7: Primary controlled CIFAR-10-C results. Regret is the mean accuracy shortfall from the per-cell oracle, on the proportion scale; lower is better. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Commitment rate is $(A + F)/n$, not interval coverage. Cells share checkpoints and residual pools.

Candidate	n	KGA	Adapt	Freeze	A/F/U	FA_u	Commitment rate
Tent	2160	0.0018	0.0080	0.1239	1094/334/732	0.0000	0.6611
EATA	2160	0.0016	0.0033	0.1313	1207/118/835	0.0000	0.6134
SAR	2160	0.0017	0.0003	0.1405	1430/0/730	0.0005	0.6620

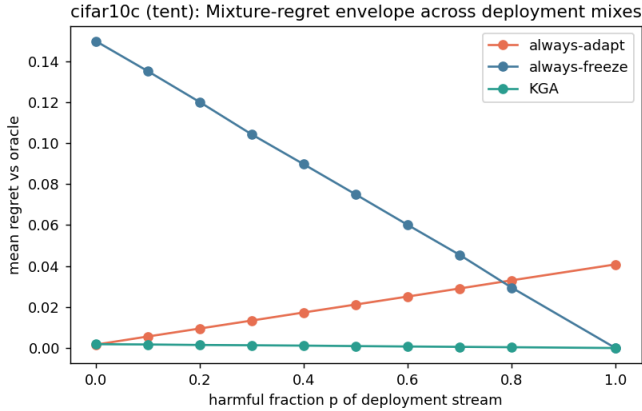


Figure 4: Historical CIFAR-10-C Tent seed-0 mixture illustration, using the preserved `stress_grid_multiseed_v1/seed0` curve. Mean regret is plotted against the constructed harmful fraction p for always-adapt, always-freeze, and KGA. The curve illustrates changing mixture weights within that saved run; it does not report the current Protocol B aggregate or independently sampled deployments.

over always-adapt (95% CI $[+0.0023, +0.0095]$) and $+0.1221$ over always-freeze (95% CI $[+0.0630, +0.1777]$). Across five stream and evaluation runs (seeds 0–4) on the archived source checkpoint, per-seed regret reductions for Tent are $+0.0073, +0.0063, +0.0058, +0.0061, +0.0055$, strictly positive across each evaluated run. These repetitions evaluate stream variation rather than variation across independently trained source models. Historical run-seed bootstrap vectors from earlier development cycles are cataloged separately in the supplement. Corruption-family sensitivity breakdowns appear in Table 34.

8.2 Observed Interval Inclusion Is a Separate Diagnostic

Commitment rate measures how often the interval excludes zero. Interval inclusion measures how often it contains the observed cell outcome. Table 8 reports these diagnostics separately under Protocol B (disjoint fit/calibrate/score cross-fitting) at nominal target $1 - \alpha = 0.90$.

Observed inclusion rates closely track the 90% nominal level across all three candidates: Tent includes 1,954 of 2,160 outcomes (90.46%), EATA includes 1,957 (90.60%), and SAR includes 1,933 (89.49%). These are observed out-of-fold inclusion rates on an opened, dependent benchmark; they do not establish interval coverage on unseen distributions. Evaluating directional errors relative to committed decisions reveals asymmetric behavior: EATA commits one false FREEZE among 118 FREEZE decisions ($FF_c = 0.0085$, occurring in Gaussian noise). Tent commits zero false FREEZE decisions among 334 FREEZE decisions ($FF_c = 0.0000$), while SAR commits zero FREEZE decisions and one false ADAPT among 1,430 ADAPT decisions ($FA_c = 0.0007$, in pixelate). These conditional denominators describe the observed directional errors among committed decisions on this panel, reflecting low empirical error rates on the evaluated records.

8.3 Synchronized Development Baseline Comparison

Table 9 compares KGA against fixed policies, scalar heuristics, and an uncalibrated point gate on 2,160 CIFAR-10-C Tent rows, using identical saved features and candidate-specific oracles under Protocol B.

KGA achieves lower point regret (0.0018, mean accuracy 0.7931) than either fixed policy (always-adapt 0.0080, always-freeze 0.1239), while committing zero false adaptations ($FA_u = 0.000$, zero false ADAPT decisions out of 1,094 ADAPT commitments). The same-prediction point gate achieves even lower point

Table 8: Retrospective interval diagnostics for the controlled cells under Protocol B (disjoint fit/calibrate/score cross-fitting), at nominal target $1 - \alpha = 0.90$. Observed inclusion concerns Δ^{cell} . Candidate/family breakdowns and directional-error denominators are provided in the supplement.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	FA_u/FA_c	FF_u/FF_c
TENT	2160	0.9000	0.9046	0.0469	0.0483	0.6611	0.0000/0.0000	0.0000/0.0000
EATA	2160	0.9000	0.9060	0.0387	0.0378	0.6134	0.0000/0.0000	0.0005/0.0085
SAR	2160	0.9000	0.8949	0.0279	0.0277	0.6620	0.0005/0.0007	0.0000/-

Table 9: Current outcome-disjoint saved-feature comparison on 2,160 CIFAR-10-C Tent rows, computed from the same paired scores. All rows have the same oracle and weights; both fitted gates use identical predictions. Lower regret is better. FA_u is the fraction of all rows with ADAPT and $B \leq 0$; commitment is $(A + F)/n$.

Policy	Regret	Mean acc.	FA_u	Commitment	ADAPT	FREEZE	ABSTAIN
Always Adapt	0.0080	0.7869	0.334	1.000	2160	0	0
Always Freeze	0.1239	0.6709	0.000	1.000	0	2160	0
Confidence increase	0.0080	0.7869	0.250	1.000	1834	326	0
Entropy decrease	0.0081	0.7868	0.249	1.000	1812	348	0
Point benefit (same prediction)	0.0004	0.7945	0.041	1.000	1471	689	0
Current KGA	0.0018	0.7931	0.000	0.661	1094	334	732

regret (0.0004, mean accuracy 0.7945) by committing to adaptation whenever $\hat{\Delta} > 0$ without an uncertainty margin; on the evaluated panel, this rule records 88 nonpositive adaptations ($\text{FA}_u = 0.041$, 88/2,160). KGA trades point regret for greater directional conservatism: by abstaining on 732 cells whose intervals contain zero, it records zero false ADAPT decisions on these evaluated cells. The scalar heuristics (confidence increase and entropy decrease) achieve regret of 0.0080 and 0.0081, respectively, with substantial false adaptations ($\text{FA}_u = 0.250$ and 0.249) and lack formal calibration. An expanded nine-policy decision rule comparison—including drift monitors, ATC-style estimators, and fixed margins—is reported in Table 25.

8.4 CCT-20: Conservative Retention, Not Selective Routing

On the recorded CCT-20 cells, KGA served the frozen model throughout and avoided the measured loss of always adapting. Across 45 evaluation cells, KGA issued 0 ADAPT, 44 FREEZE, and 1 ABSTAIN decisions. Because every non-adapt decision serves the base model, KGA served frozen predictions across the entire benchmark.

KGA matched always-freeze and avoided the measured degradation of always-adapt. The predeclared utility-retention endpoint was therefore met. The stronger selective-routing criterion was not: KGA issued no ADAPT decisions and failed to select the single helpful candidate. All recorded development

cells for Tent had negative measured benefit.

A separate retrospective labeled finite-frame audit is reported in Appendix S. That audit uses a different interval construction and produces 0 ADAPT / 35 FREEZE / 10 ABSTAIN actions; these are audit-assisted decisions and do not revise the primary KGA actions reported here (0 ADAPT / 44 FREEZE / 1 ABSTAIN).

8.5 Development Stops and Adverse Diagnostics

The earlier v1 study on So2Sat-LCZ42 [28] tested 2 candidates on 9 development cities and 5 independent checkpoints. Tent had both helpful and harmful cities, but neither candidate passed all predeclared eligibility checks. The study stopped before gate calibration: no target inputs or labels were accessed, and there is no target natural-shift score. The development outcomes are now open, so retuning them cannot produce fresh confirmation. The separately locked v2 calibration study is reported in Section 8.9; it also stopped before target access, after failing its fixed calibration eligibility screen.

The auxiliary studies also show limitations. ImageNet-R and PACS favor always-adapt; ImageNet-C is sensitive to the candidate and its operating point. Office-Home and RxRx1 mainly retain the frozen model. The opened one-sided Camelyon17 panel adapts throughout because all 18 cells are helpful.

Table 10: CCT-20 utility-retention endpoint defined in Eq. (12). Contrasts are baseline regret minus KGA regret on the set-membership accuracy-proportion scale. Contrasts report nominal pointwise 95% two-way bootstrap intervals. Location-level records are detailed in the supplement.

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

A retrospective 72-condition Camelyon17 diagnostic contains both helpful and harmful candidate effects but issues 0 ADAPT, 2 FREEZE, and 70 ABSTAIN decisions, matching always-freeze regret (0.0438; Appendix B.5). Together with CCT-20 and RxRx1, it provides retention evidence without ADAPT exposure. These studies do not establish prospective selective routing on unseen natural shifts.

8.6 Additional Tests of Decision Value

The following analyses ask three related questions at different levels. First, does residual calibration add decision value beyond a shared point predictor? Second, can paired multiclass transport make a decision under an explicitly declared structural assumption? Third, does a natural-shift calibration screen provide enough evidence to authorize target scoring? Sections 8.7–8.9 examine these questions separately; Section 8.10 describes the local deployment integration. None turns a conditional or retrospective result into a general deployment guarantee.

8.7 What Does Calibration Add to a Shared Predictor?

We conducted a new retrospective comparison on the already opened CIFAR, ImageNet, and mixed-checkpoint records. Every arm shares the same fitted GBRT within a fold. CIFAR and ImageNet folds exclude complete scored corruption types from fitting, tuning, and calibration. Checkpoint-only folds exclude the scored source model and rotate held-out semantic groups; joint folds exclude both the scored source model and the complete corruption type. This tests transfer beyond the earlier cell-wise splits, while retaining the historical data’s provenance and dependence limits. It is not a new independent deployment study. The loss grid, fold construction, and policy family were fixed before the new fitting; held-out outcomes enter scoring only after decisions have been saved.

The primary loss uses a prespecified harm weight

of five:

$$L_5 = \frac{1}{N} \sum_{i=1}^N [(1 - a_i)(B_i)_+ + 5a_i(-B_i)_+], \quad (14)$$

where B_i is measured paired cell benefit and a_i is one for ADAPT and zero otherwise. A fixed margin minimizes this loss on tune-only groups. Constant and locally scaled residual intervals use the exact conformal rank, without clipping a rank that exceeds the calibration sample size. We report $100L_5$ in percentage points, so lower values indicate lower weighted decision loss. The full study includes harm weights 1, 5, 20, nominal levels 80, 90, 95%, group-maximum residuals, exposure-matched controls, and binomial risk-selection arms.

Table 11 does not establish a general advantage for calibration. Scaled calibration reduces CIFAR Tent loss from 0.2867 to 0.1429 percentage points versus point prediction, and from 0.2264 versus the tuned margin. Its loss at matched exposure is only modestly lower than point ranking (0.1429 versus 0.1610). Simple margins do better on CIFAR EATA and SAR; point or margin policies do better in the other cohorts. Within a fixed scoring fold, subtracting the same finite radius from the same point predictions preserves their ordering. At matched ADAPT exposure, the constant-radius rule selects the same cells as the point-ranking control, using the same tie rule. This does not compare rankings across separately fitted folds or studies; locally varying radii can change the ordering. The favorable Tent comparison is a point estimate on six observed corruption types. Omitting JPEG only as an influence diagnostic nearly removes the advantage over point prediction and reverses the comparison with the tuned margin; all six corruption types remain in the primary estimate. Paired bootstrap ranges across these types describe sensitivity, not a population confidence interval or multiplicity-adjusted discovery (Appendix R).

Nominal 90% constant-cell intervals include only 50.51–60.69% of held-out CIFAR benefits; scaling and group maxima do not restore nominal inclusion. Decision loss and interval inclusion answer different

Table 11: Retrospective shared-predictor comparison. Losses are $100L_5$ in percentage points (lower is better). Margin is selected on tuning data; Constant and Scaled use nominal 90% cell residual intervals. Inclusion C/S is observed held-out interval inclusion for those two arms, not a coverage guarantee. The same 108 mixed-checkpoint records appear in two evaluation designs; they are not 216 independent observations.

Cohort / outer hold-out	Cells	Point	Margin	Constant	Scaled	Inclusion C	Inclusion S
CIFAR EATA / corruption	2160	0.1077	0.0663	0.1556	0.0914	54.35%	51.67%
CIFAR SAR / corruption	2160	0.1677	0.0465	0.1291	0.0844	50.51%	48.19%
CIFAR Tent / corruption	2160	0.2867	0.2264	0.2150	0.1429	60.69%	58.24%
ImageNet EATA / corruption	135	0.0481	0.4443	0.6939	1.8787	91.11%	92.59%
ImageNet SAR / corruption	135	0.0000	0.2115	2.5989	1.8754	97.04%	93.33%
ImageNet Tent / corruption	135	2.1913	1.3756	1.4465	1.4465	93.33%	96.30%
Mixed Tent / checkpoint	108	0.3236	0.4032	0.7954	5.2708	85.19%	90.74%
Mixed Tent / checkpoint + corruption	108	0.5829	0.4611	0.9403	2.6329	79.63%	87.96%

questions: a lower observed loss does not establish nominal coverage, and observed inclusion does not establish useful routing. All independently held-out environment-calibration arms abstain: this split provides no calibration environments independently excluded from fitting and tuning. The 10% conditional group-risk arm also abstains throughout. Even with zero observed errors, its 13-threshold family needs 53 exposed calibration groups at within-fold confidence error .05, exceeding the largest partition of 45. These are information and transfer limits, rather than evidence that every possible policy is unsafe. The 20% sensitivity selects some CIFAR updates; its held-out group error rates differ materially from cell rates (Appendix O). B6 remains descriptive: this study does not recover its missing original execution identities or repair its historical rank clipping.

8.8 A Conditional Paired-Transport Certificate

The population frontier leaves open how extra structural information can make a decision possible. We instantiate one such assumption regime for fixed multiclass predictors. Let $r = (f_0(X), f_a(X))$ index their joint prediction bins, $A_{ry} = P_S(r \mid Y = y)$, and $q_r = P_T(r)$. With unknown target prior π , declare

$$\sum_y \pi_y \text{TV}(P_T(r \mid y), A_{.y}) \leq \rho. \quad (15)$$

The budget ρ requires external justification; unlabeled evidence cannot generally validate it. Both predictors must be fixed independently of the source and target evidence samples. In particular, adapting on the same target examples later used as independent evidence does not satisfy this construction.

For R bins and K classes, simultaneous binomial

intervals for the $RK + R$ source-conditional and target-marginal probabilities define a linear feasible set. The variables are target joint masses t_{ry} , transported source masses $s_{ry} = A_{ry}\pi_y$, priors, and absolute-value slacks for Equation (15). Source interval endpoints multiply π_y , so the resulting constraints remain linear. Minimize and maximize

$$\Delta(t) = \sum_{r,y} (\mathbf{1}\{f_a(r) = y\} - \mathbf{1}\{f_0(r) = y\})t_{ry}. \quad (16)$$

On the simultaneous interval event, the true table is feasible and these extrema enclose its benefit. Under the stated sampling and transport assumptions, strict interval thresholding therefore has unconditional wrong-direction probability at most α . It does not give acceptance-conditional or repeated-use error control. Detected infeasibility, solver failure, or an interval touching zero produces ABSTAIN. The full constraints, proof, and numerical scope appear in Appendix P.

This construction applies existing interval-constrained LP inference [57] to a paired contrast. Joint optimization can retain information lost by separately bounding the two accuracies: in one exact three-class construction, the paired interval is $[.1, .4]$, whereas subtracting separate extrema on the same feasible set gives $[-.1, .6]$. An unidentifiable prior can therefore coexist with an identified positive action. This is a concrete use of dependence, not a first claim for action identification [56] or new LP machinery.

The paired procedure adapts in all 60 cancellation trials while the separate accuracy relaxation always abstains. Near a small positive benefit, it avoids the plug-in rule’s 22 wrong FREEZE decisions out of 60 at the smaller sample size; its price is abstention. Larger sensitivity budgets likewise reduce commitment. Most importantly, the same observed evidence

Table 12: Selected diagnostic regimes from 4,860 synthetic trial configurations (nine scenarios, three sample sizes, three sensitivity budgets, 60 trials each). A/F/U count ADAPT/FREEZE/ABSTAIN over 60 trials. Separate subtracts accuracy extrema on the same polytope; it is not an implementation of MaC-LP. The same-evidence violation row deliberately violates the declared assumption and reuses the paired-cancellation row’s observable count streams. These archived outputs retain the earlier numerical qualification described in Appendix P.

Synthetic regime	n/class	ρ	True Δ	Paired A/F/U	Separate A/F/U	Plug-in A/F/U
Paired cancellation	100	0.0	0.16	60/0/0	0/0/60	60/0/0
Same evidence, violated assumption	100	0.0	-0.50	60/0/0	0/0/60	60/0/0
Near null	100	0.0	0.02	0/0/60	0/0/60	38/22/0
Near null	10000	0.0	0.02	20/0/40	0/0/60	60/0/0
Helpful	1000	0.0	0.20	60/0/0	60/0/0	60/0/0
Helpful, wider sensitivity	1000	0.1	0.20	0/0/60	0/0/60	60/0/0
Harmful	1000	0.0	-0.20	0/60/0	0/60/0	0/60/0

can have true benefit $-.5$ under a violated transport assumption: the paired procedure then makes 60 harmful ADAPT decisions. The diagnostic demonstrates both conditional value and the impossibility of obtaining the missing assumption from unchanged unlabeled observations. It does not establish useful natural-shift adaptation.

8.9 Natural-Shift Calibration and Execution Costs

The separate So2Sat v2 study fixes the controller derived from nine already opened development cities and uses five independently trained source checkpoints. Its calibration unit is a city: the score is the maximum absolute residual across all five checkpoints, with the 18th order statistic among 19 city scores at $\alpha = .10$. Under exchangeability, this construction targets simultaneous coverage of the five checkpoints for one new city. It does not give simultaneous coverage of ten target cities, coverage conditional on passing an eligibility screen, or a demonstrated guarantee under the held-out geographic shift. The chronology documents the study sequence but does not establish general prospective validity.

The locked calibration completed all 95 city-checkpoint cells. Helpful and harmful development effects were both present, but the uncertainty interval was too wide to meet the pre-target action-exposure criterion. Its radius was 0.14224 (14.224 percentage points), and its decisions were 2 ADAPT, 0 direct FREEZE, and 93 ABSTAIN. Both accepted updates were helpful in the calibration records, but they occurred in only one city; no city supplied a direct FREEZE. Thus both prespecified requirements of at least seven direct-action cities failed, and the study

stopped with zero target pixels or labels read. No target authorization or target score was produced. The observed mean calibration utility was 0.0989 percentage points for the gate, 0.1939 for always-adapt, 0.1264 for the checkpoint-only point baseline, and zero for always-freeze. These same calibration outcomes set the radius and screen, so these utilities are descriptive and cannot demonstrate independent target benefit.

A posthoc diagnosis of those opened records found 51 helpful, 43 harmful, and one tied candidate effect. The radius was set by São Paulo checkpoint 0: its predicted benefit was 14.4215 percentage points but its observed benefit was only 0.1976 points. The resulting radius exceeded every negative prediction magnitude (at most 2.6067 points), preventing direct FREEZE. Thus the failed screen reflects insufficient prediction accuracy and interval informativeness, rather than an absence of helpful and harmful candidates. This diagnosis did not refit the controller, alter the screen, or authorize target access.

Calibration took 36.55 minutes. The controller requires five shadow candidates before routing a city, so retaining the frozen model does not avoid candidate-construction cost. In a separate profile on one opened development city, that process took a median 207.44 seconds, excluding subsequent serving inference and cleanup. These measurements are workload-specific; calibration acquisition, energy, and long-duration maintenance costs remain unmeasured. Stage timings, memory measurements, and measurement conditions are reported in Appendix J.1.

8.10 A Local Deployment Integration

We also implemented a local fail-closed deployment lifecycle that isolates the candidate from the active checkpoint, rejects malformed or outcome-bearing evidence, and retains the frozen model when assessment fails. The implementation records calibration and model identities and exercises rollback and persistence failures. These tests evaluate software behavior, not the statistical validity of the certificate or field reliability. Detailed fault tests, integrity records, and workload-specific latency measurements are reported in Appendix J.1.

9 Limitations and Broader Impact

9.1 Assumptions, Not an Unconditional Safety Guarantee

The population frontier is exact within its declared rich binary class. A credible residual bound must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions, while an unsupported bound cannot justify a strict population claim. The useful object is the class-dependent feasible set, not the algebraic inequality alone.

Justifying the population residual bound. A development-domain proxy does not resolve the population residual bound required by Theorem 3. Within the declared class, β must come from external domain knowledge, target labels, or an explicit transfer assumption. Processing the same label-free evidence cannot supply that missing restriction. The supplementary discussion records the withdrawn proxy analysis (Appendix B.11).

For KGA, empirical interval coverage requires separate support at the intended decision unit. A larger radius reduces commitment but can increase regret when the frozen model is worse. A smaller radius increases exposure and makes calibration failures more consequential. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem and requires recalibration or a justified transfer argument. The new held-out-corruption comparison directly exhibits this failure: nominal 90% calibration transfers poorly on CIFAR, and the best useful policy depends on the candidate and harm weight. The paired-transport certificate supplies a different sufficient assumption regime; it does not validate either that calibration transfer or its own

transport budget from unlabeled target data.

Repeated deployment needs a separate guarantee. A predeclared policy does not by itself supply exchangeability or time-uniform safety. Reusing calibration data, changing the frozen baseline after accepted updates, and selecting later candidates from earlier outcomes change the joint process. Adaptive conformal inference studies online coverage frequency with feedback [39]; that is not time-uniform control of KGA’s false commitments. Selection-conditional and group-conditional guarantees likewise require additional methods, as discussed in Section 2.

9.2 What the Experiments Establish—and What Remains Open

Four limitations determine how these experiments should be interpreted:

1. **Retention vs. Opportunistic Routing under Natural Shift:** On real-camera distribution shifts (CCT-20), KGA served the frozen model across all 45 cells and avoided the measured degradation of always-adapt in that study. However, it made zero ADAPT decisions. Prospectively demonstrating selective opportunistic routing on an unopened natural-shift target remains an open empirical challenge. The retrospective Camelyon17 diagnostic in Appendix B.5 similarly exhibits conservative retention (zero ADAPT decisions under an outcome-blind decision rule).
2. **Retrospective Benchmarks vs. Prospective Deployment:** The controlled CIFAR-10-C and ImageNet-C experiments evaluate retrospective, opened panels under their track-specific declared protocols (including Protocol A and Protocol B). While these report empirical commitment and inclusion diagnostics on opened records, prospective evaluation on unopened environments is required to assess distribution-transfer properties.
3. **Checkpoint and architecture transfer.** The primary CIFAR-10-C panel uses one source checkpoint. Additional checkpoint comparisons are descriptive: B6’s clipped calibration rank does not establish nominal coverage, and incomplete execution provenance limits independent replication claims. Prospective transfer to unseen model families and environments remains untested (Appendix B.3.3).
4. **External method comparisons.** The shared-feature baselines and scoped native components isolate specific decision rules, but do not con-

stitute complete reproductions of all published learners or a full official-method comparison (Appendix B.3.4).

The controlled studies quantify candidate-specific utility and commitment tradeoffs. CCT-20 provides retention evidence without ADAPT exposure. These experiments do not establish population coverage or protection on unseen environments. A unified statistical synthesis matrix consolidating sample sizes, evaluation units, nominal confidence intervals, multiplicity adjustments, and verdicts across the principal empirical tracks summarized here is reported in Table 44.

9.3 Operational Scope and Potential Harm

A rejected update can prevent adaptation harm, but retaining a frozen model is not proof that the model is accurate or suitable for a high-stakes setting. Misinterpreting an abstention as a certificate would be particularly risky. In high-stakes applications, an unresolved assessment calls for an application-specific fallback, with domain-expert review where appropriate; the choice depends on the consequences of continuing with the frozen model. The reported experiments do not validate clinical, safety-critical, or autonomous operation. No held-out physical camera deployment or common end-to-end latency result is claimed.

An unavailable assessment must remain distinct from evidence of negative benefit: both retain the frozen model, but only the latter supports a strict FREEZE decision. The local integration checks this behavior; its tests do not establish calibration validity or field reliability. Detailed implementation and reproducibility information is provided in the supplement.

10 Conclusion

The update decision depends on what the available evidence can distinguish. For the declared binary target class, K-Bound identifies the feasible population-benefit interval and its exact strict-decision frontier. The evidence-fibre audit establishes the accompanying limit: unchanged label-free observations cannot justify an arbitrarily tighter residual bound. The paired-transport construction illustrates a separate multiclass route under an explicit conditional-shift budget and sampling assumptions; its usefulness depends on the credibility of those assumptions.

KGA makes the empirical question concrete by predicting paired evaluation-cell benefit and calibrating an interval around that prediction. Its value is candidate- and operating-point-dependent. Controlled routing gains coexist with simpler-rule wins. In the shared-predictor comparison, scaled calibration has lower observed CIFAR Tent harm-weighted loss, but the advantage narrows at matched exposure and depends on the corruption mix. Nominal 90% constant-cell inclusion falls to 50.51–60.69% on held-out CIFAR corruptions. Useful decisions and reliable intervals must therefore be evaluated separately.

Natural-shift evidence is more limited: CCT-20 demonstrates retention without ADAPT exposure, and So2Sat stops before target scoring. The present studies do not establish population-risk protection or useful selective routing on unseen natural environments. The next empirical test is whether independently supported calibration can accept beneficial updates and reject harmful ones under a fixed protocol. Where the available information cannot support either strict sign, abstention records the unresolved decision; it does not certify that the retained model is safe.

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